

EXAM 7 – SPRING 2014

12. (1.75 points)

A single retrospectively rated policy has the following adjustments:

<u>Adjustment</u>	<u>Incremental Losses</u>	<u>Incremental Premium</u>
1	\$100,000	\$150,000
2	50,000	25,000
3	15,000	0
4	20,000	5,000
5	5,000	-1,250

a. (0.5 point)

Discuss why incremental premium can be greater than incremental losses at the first adjustment.

b. (0.5 point)

Explain how a positive fourth adjustment to the incremental premium could follow a third adjustment of \$0.

c. (0.75 point)

Explain how the observed premium development to loss development (PDL) ratio could become negative for the fifth adjustment.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 12	
TOTAL POINT VALUE: 1.75	LEARNING OBJECTIVE: A14: Forecast Premium Reserves.
SAMPLE ANSWERS	
Part a: 0.5 point	
Several reasons were accepted. For full credit, the candidate could either list and briefly describe at least 2 items, or list 1 item with a more detailed explanation. Partial credit was provided for only listing 1 item without a detailed explanation. We accepted: - Tax Multiplier - Loss Conversion Factor - Basic Premium - Fact that few of the losses are already capped by the first adjustment period <i>Sample Answer 1</i> Premium at the first adjustment includes the basic premium, which is the amount charged even if no losses emerge. Also, losses are not capped as much at the first adjustment, so multiplying by the loss conversion factor and tax multiplier can cause the PDLD ratio to be greater than 1.0. <i>Sample Answer 2</i> In the first incremental period the premium includes an amount related to expenses and insurance charges (Basic Premium) in excess of the amount driven by losses. The effects of the Tax Multiplier and Loss Conversion Factor also contribute to the incremental premium being greater than the incremental losses at this point.	
Part b: 0.5 point	
We awarded credit for describing what could be causing 0 incremental premium at the 3rd adjustment, and also for explaining what would have been different about the 4th adjustment to cause the positive incremental premium. The most common answer was to say that the loss development at the 3rd adjustment was on a loss that had already reached the per accident limit, and the loss at the 4th adjustment was either on a new claim or one that had not yet reached the limit. We also accepted alternative solutions, including changes to the terms of the policy. <i>Sample Answer 1</i> It is possible that the 15,000 incremental loss in the 3rd adjustment is for a claim that has already breached the per-accident limit, so it generates 0 capped loss at the 3rd adjustment, and thus 0 incremental premium.	

SAMPLE ANSWERS AND EXAMINER'S REPORT

The loss in the 4th adjustment could be from another claim which has not breached the per-accident limit, and so it will generate incremental premium.

Sample Answer 2

If all development in the third period occurred out of the loss limits, it is possible for the incremental premium to equal 0. For the incremental premium to be greater than 0 in the 4th period, it means that the development in the 4th period was within the loss limits.

Part c: 0.75 point

There were 3 main points that were needed to get full credit.

- Losses on claims below the cap decreased
- Losses on claims above the cap increased
- The increase in loss above the cap was greater than the decrease in loss below the cap, which caused the incremental premium to drop despite the fact that incremental losses were positive.

Sample Answer 1

Some incremental loss amount greater than 5k that was already outside plan parameters offset by slightly lesser negative incurred development on claims still inside the plan parameters, which would bring negative premium development. In short, losses in total increased but capped losses decreased.

Sample Answer 2

By the 5th adjustment, we may be seeing positive development only in losses that have been capped already, whereas claims at lower levels may be settling at amounts below what had previously been reserved. Therefore there is upward loss development outside of loss limits, and downward, but lower, loss development within limits. This would cause negative premium for the loss within the cap and no additional premium for loss above the cap.

EXAMINER'S REPORT

Candidates struggled a bit with this question and ran into trouble when they did not provide complete explanations, or where they ignored details given in the problem (by providing explanations that were not possible based on the numbers given).

Part a

Candidates did best on this part of the question. Candidates needed to know how the incremental premium is calculated at the 1st adjustment, and be able to explain how portions of that calculation could cause the PDLD ratio to be above 1.0 at that point. To get full credit, the candidate needed to either go into detail on one of the potential reasons, or briefly describe more than one. The most common error was just naming one element of the formula, without discussing what it is or why it contributes to PDLD ratio being greater than 1.0.

SAMPLE ANSWERS AND EXAMINER'S REPORT

Part b

Candidates struggled a bit on this part. Candidates needed to understand the various limits that could prevent the incremental losses from influencing the premium. To get full credit, the candidate needed to provide an explanation both of what could be causing the phenomenon at the 3rd adjustment, and what would have changed about the 4th adjustment. The most common error was not explaining the 4th adjustment at all.

Part c

Candidates performed worst on this part. The required understanding for this part was very similar to what was needed for part b. To get full credit, the candidate needed to explain which types of loss (relative to the caps) were increasing, and which were decreasing. They also needed to point out the relationship between the two types (which was greater). One of the more common errors was not pointing out that the increases had to outweigh the decreases. We accepted many ways of stating or even implying this relationship, but some candidates did not point it out at all. Other candidates ignored the caps completely, only stating that "losses went down, so premium was returned."